

Interactive Digital Art and Design SE608

Project Proposal Year IV



Student Name:	Yullia Antonova	Student Number:	C00301432
Working Title:	Interactive Generative Digital Plant		
Description:	This project will create an interactive digital plant using generative art. The plant will grow and change in response to the user's interaction. Different parts of the plant, such as the stem, leaves and flowers, will be generated digitally and will change their shape, movement and appearance. The aim is to create a visually interesting interactive experience where the user can influence the growth and behaviour of the digital plant.		
Reasons for selecting project:	I chose this project because I am interested in digital art, animation and interactive experiences. I would like to explore how a simple natural object, such as a plant, can be recreated as a digital interactive experience. This project will allow me to improve my skills in creative coding, animation and generative art. I also want to experiment with creating a system where the visual result is not completely fixed, but changes according to the user's interaction. I think this project will help me combine my artistic and technical skills and create something visually interesting and interactive.		
Proposed research content:	I will research generative art, interactive digital art and animation. I want to learn how code can be used to create a digital plant and control how it grows and moves. I will research how stems, leaves and flowers can be created using code. I will also explore how the user can interact with the plant and change its behaviour. I want to find out how code and interaction can make a digital plant look more natural, dynamic and alive.		

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External links (if applicable):	p5.js Examples https://p5js.org/examples/ A collection of examples showing animation, user interaction, generative graphics, 3D objects and procedural techniques in p5.js. I will use these examples to research how code can control movement and interaction in my digital plant. Magic Ivy – Interactive 3D Plant https://sites2.org.aalto.fi/medialabdemodays_old/projects/magic-ivy/ An interactive and generative 3D plant created using JavaScript and Three.js. The plant is generated with code and responds to user interaction. I will use this project as a reference for combining procedural plant growth with interactive 3D visuals. BOTanic Garden – A p5.js Generative Artwork https://www.behance.net/gallery/76353549/BOTanic-Garden-A-p5js-generative-artwork A generative plant simulation created with p5.js. The plants grow, reproduce and die according to programmed rules, creating a changing digital environment. I will research this project to understand how natural plant growth can be represented using code. All A Garden – Interactive Art https://nesrasportfolio.com/all_garden An interactive botanical artwork created with p5.js. User movement affects the growth of flowers and thorns. I will use this project as a reference for exploring how user interaction can change the behaviour and appearance of digital plants.
Hardware requirements:	Computer with keyboard and mouse.
Software requirements:	p5.js, JavaScript, HTML, CSS and Adobe Photoshop
Other requirements:	Internet access for research and testing.

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Signed: Yuliia Antonova

Date: 16.09.2026

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Approved/Not Approved:	
Reasons for not approving project:	
Conditions attached to approving project:	
Approved/Not Approved:	
Name of Supervisor:	
Signed: _____	Date: _____